Department of Computer Science

Faculty of Computing

UNIVERSITI TEKNOLOGI MALAYSIA

PROJECT TITLE :	SMART CAMPUS DATA SOLUTION - IMPROVING STUDENT EXPERIENCE THROUGH DATA
SUBJECT CODE/COURSE NAME :	SECP1513/TECHNOLOGY & INFORMATION SYSTEM
SEMESTER :	Sem 1 – 2025 2026
GROUP MEMBERS :	1. Name : QISTINA BATRISYIA BINTI NOOR MOHD AZLAN *Matric No : A25CS0340* 2. Name : MURSYIDAH BINTI JAHIDI *Matric No : A25CS0286* 3. Name : NOR AIN FAHIRA BINTI MUHAMAD FARIQ *Matric No : A25CS0295*

1.0 Introduction

Attendance plays an important role in students' academic performance and reflects their discipline. However, some attendance issues often occur among students.

This project aims to improve the current attendance system's efficiency. Moreover, teamwork experience and practical knowledge of application in real-life situations are gained by the team.

2.0 Detailed step & description in design thinking

2.1 Empathy

The empathy phase is to understand the users' attendance challenges faced. Data were collected via Google Form survey to identify their needs and experiences.

2.2 Define

The define phase focuses on analysing data from the previous phase to identify users' challenges. The issues of the attendance system were defined to guide the suitable solution development.

2.3 Ideate

The ideate phase aims to generate ideas based on problems defined. Brainstorming sessions were conducted to generate improvements to the attendance system.

2.4 Log Journal

This log journal outlines weekly activities conducted throughout the project process. Each activity was completed as the project timeline to ensure smooth progress. Weekly project activities are summarised in Appendix B.

3.0 Detailed description

3.1 Problem

After conducting the survey, several problems were identified. These include unnecessary attendance taking effort, poor mobile experience, inconsistent system performance, device dependency, and data and feedback issues.

3.2 Solution

Ideas were brainstormed to address previous issues. Suggestions were discussed and evaluated by strengths and weaknesses. The team selected one final solution that uses a large display screen with biometric devices for physical classes and new features in the UTMSmart app for online classes.

3.3 Teamwork

Physical meetings and discussions via Telegram group to monitor workflow and progress. Team members regularly updated in the group for mutual support. Each member contributed actively to complete the Design Thinking project.

4.0 Design thinking evidence

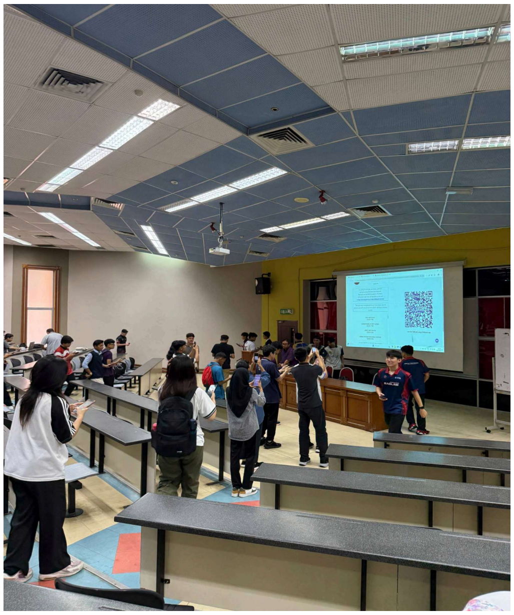


Figure 1

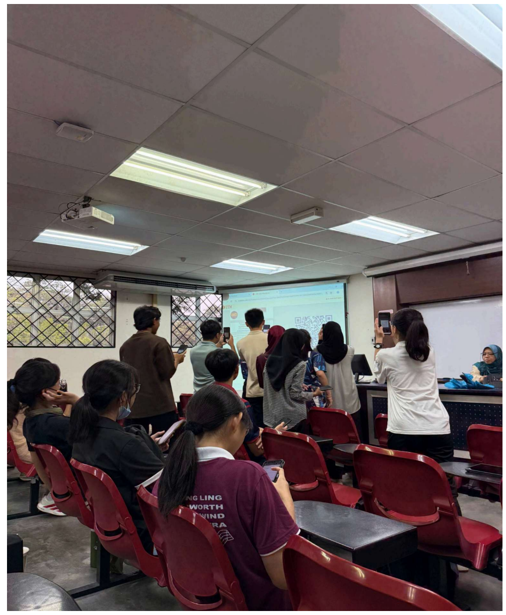


Figure 2

Figure 1 and Figure 2 shows observation evidence of QR scanning difficulties and slow internet causing frustration.

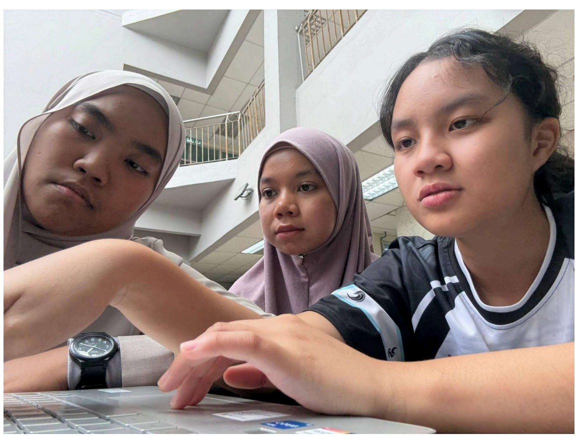


Figure 3

Figure 4

Figure 3 and Figure 4 shows meetings conducted for project development, progressing through the phases including empathy, define, ideate, prototype and testing.

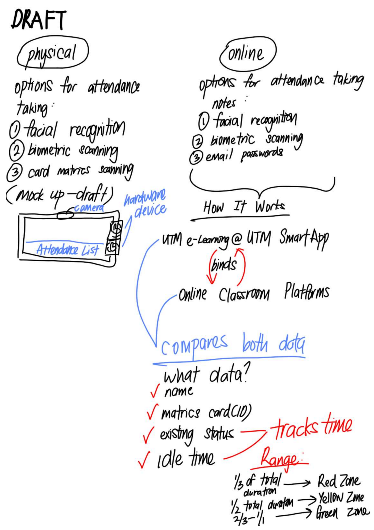


Figure 5

As problems are identified,focus is placed on the solution to overcome the current digital attendance. Figure 5 shows the prototype in its early stages.

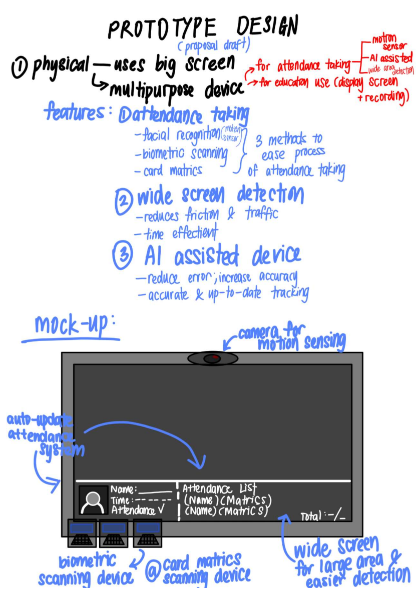


Figure 6

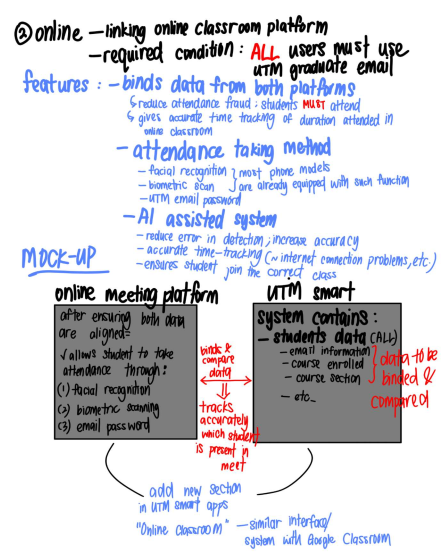


Figure 7

The final prototype draft shown in Figure 6 and Figure 7 has been designed to overcome flaws in the current digital attendance system.

4.1 Empathy

4.1.1 Data Collection Method

- Google form survey completed by 13 respondents and distributed through WhatsApp groups among UTM Students.
- Users were observed using UTMSmart to scan QR code as class starts.

4.1.2 Respondent's Profile

Figure 1 shows respondent's profiles by year of study and faculty based on data obtained from Google form survey.

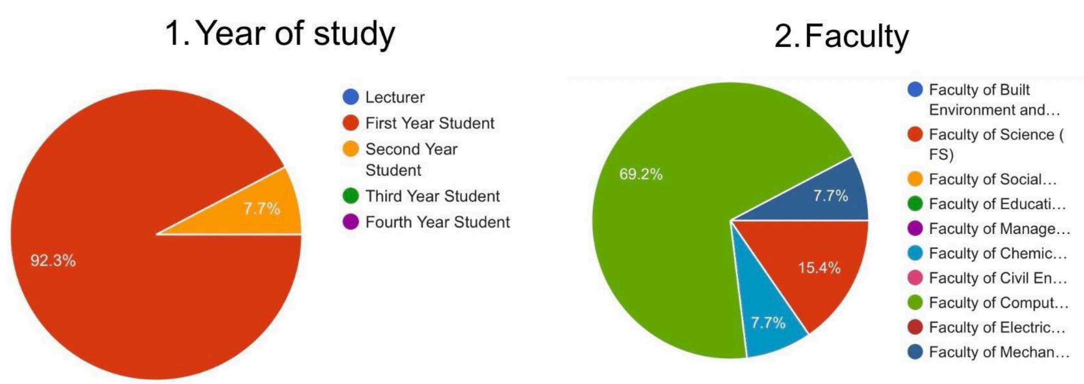


Figure 8. Respondent's profile

4.1.2 List of Possible Questions and Answers

Appendix C summarizes the list of possible questions and answers based on Google form surveys and observations.

4.1.3 Composite Character

Student A represents 13 UTM students using QR code attendance via smartphones. QR detection failure, poor internet, invalid QR code, and QR displayed only at the start are the issues faced. Time wasted, attendance disrupted and emotions are triggered by these situations.

4.2 Define

4.2.1 Problem of the existing system

Several problems were identified in the existing attendance system.

Poor mobile experience hinders QR code scanning. Low quality cameras on some devices makes scanning difficult, requiring direct scans from the lecture's screen.

Moreover, device dependency poses further concerns. Mobile devices are heavily relied upon for attendance. No alternative method exists if a student's phone battery runs out.

Lastly, data and feedback reliability remains the main problem. Attendance records may reflect actual class participation, as remote recording is possible. This encourages dishonesty as requirements are met without attending classes.

4.2.2 Persona

Details

Name : Student B

Age : 19 years old

Occupation : Undergraduate student at Universiti Teknologi Malaysia (UTM)

Background

Student B attends both physical and online classes. However, the attendance method differs depending on the lecturer. Technical issues are commonly faced by Student B.

Goals

- Attendance is recorded quickly and accurately
- Unfairly being marked absent is avoided
- A simple and reliable system is provided

User issues

- Different attendance systems used by different lecturers
- QR codes are not clearly displayed
- Slow internet causes submissions failures and invalid attendance

User needs

- A standardised digital attendance system
- Real time confirmation after submission
- Low data usage and unstable connection are supported
-

4.2.3 Journey Map

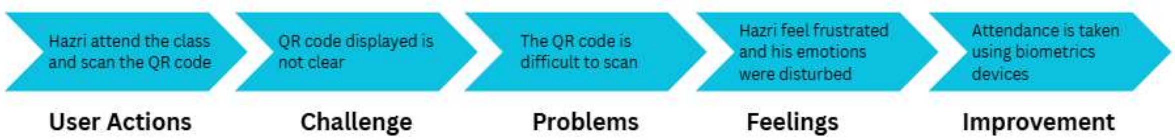


Figure 9

Figure 9 represents the student's experience during attendance taking. Initially, students attend classes and attempt QR code scanning for attendance. However, QR code scanning can be difficult and frustrating when it is unclear. Thus, an improved biometric attendance method is proposed for more reliability.

4.3 Ideate

The ideate phase includes the process of understanding flaws in the current attendance system and finding efficient solutions to be implemented.

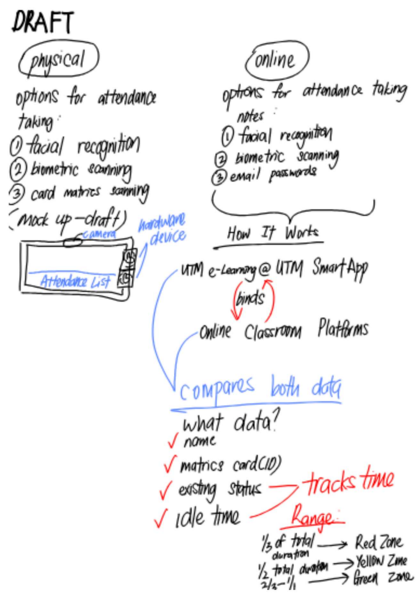


Figure 10

Initially, the prototype designed by the team in Figure 10 consisted of solutions for online and physical attendance taking where flaws of both prototypes are complemented.

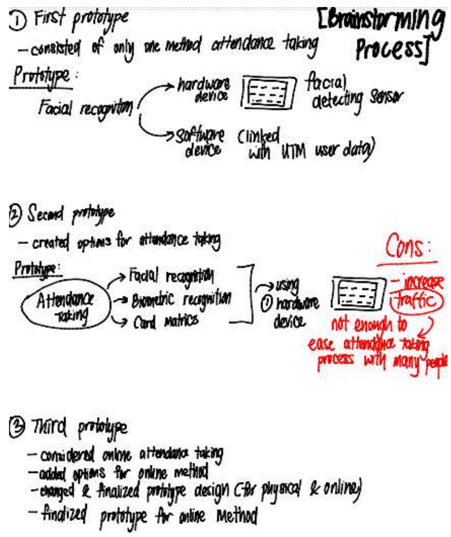


Figure 11

The prototype was improved to maximize efficiency and flow of attendance where effectively smoothen the process of attendance taking as shown in Figure 11.

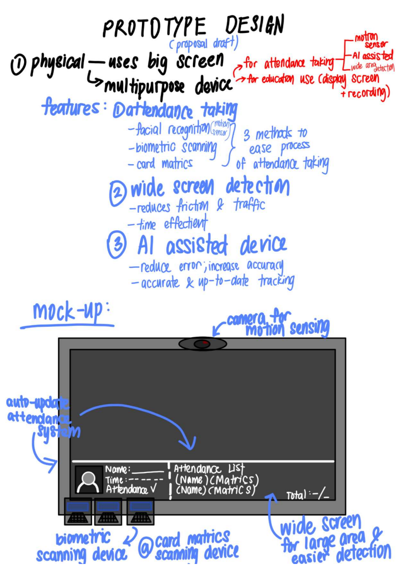


Figure 12

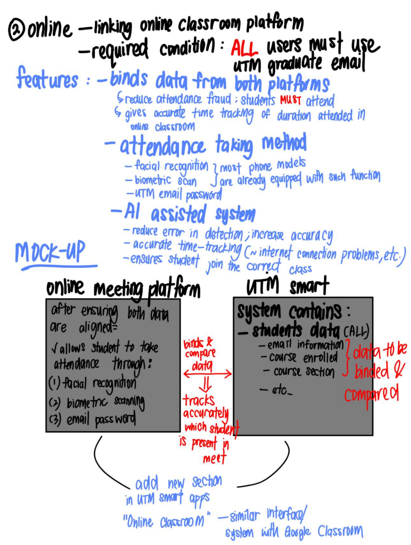


Figure 13

The prototype has been finalized with the team's agreement where problems faced by the current system are fully complemented.

4.4 Prototype

4.4.1 Physical Prototype

Figure 14 shows a physical prototype developed with a wide screen TV, motion detection camera, and fingerprint scanner. These devices are aligned to update in e-Learning and UTMSmart.

Figure 14. Physical prototype

- Wide Screen Television

  The screen serves both educational and attendance purposes. Student's names display on the screen upon successful face detection.

- Motion Detection Camera

  Multiple student faces are detected in a single frame via automatic zoom and angle adjustments. AI features allow accurate detection with bounding box coordinate mapping.

- Fingerprint Scanner

  Fingerprint scanning acts as backup in case face recognition fails due to poor lighting or seating distance.

4.4.2 Online Prototype

The next prototype addresses student attendance frauds. It integrates UTM database systems, UTM Smart Apps and online meeting platforms like Webex. The purpose of all included aspects is listed in Appendix A.

The prototype is implemented to bind data between UTM database server and online meeting platform via meeting link entry into the prototype. Then, datasets from UTM database and online meeting platforms are compared. Attendance is allowed when datasets match. Otherwise, attendance taken is invalidated.

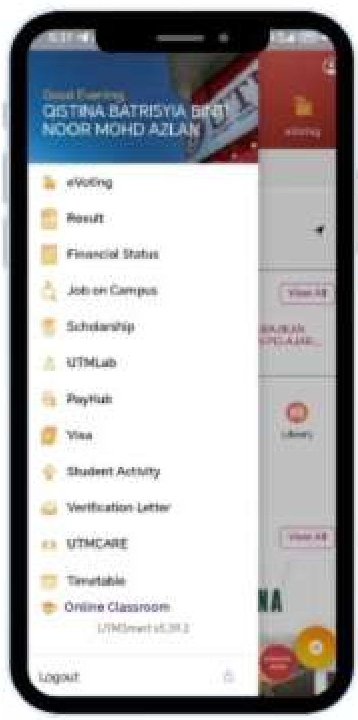


Figure 15

Figure 16

In Figure 15 and Figure 16, a new section is added by the prototype to UTM Smart Apps main interface for easy and direct student access to the online attendance page.

Figure 17

Figure 18

In Figure 17 and Figure 18, the newly added sections including “Status”, ”Duration” and “Time” were added to the prototype. Proof of attendance is showcased to mark students’ successful attendance.

4.5 Testing

Feedback of the prototype was collected from two UTM students via WhatsApp survey. Prototype was deemed more effective than QR code attendance due to automatic identification, fraud minimisation, and efficient AI and biometric data storage. However, concerns regarding late handling fairness and unclear duration tracking were identified. Further refinement is required to overcome the weaknesses.

5.0 REFLECTION

Mursyidah binti Jahidi :

Graduating with excellent grades and securing a career are my goals. Critical thinking skills are improved by handling real problems with application of design thinking. In the field of Data Engineering, this skill is essential in the industry. Problem solving skills are planned to be strengthened through the implementation of design thinking in every project to enhance industry potential.

Nor Ain Fahira binti Muhamad Fariq :

My goals for my course is to graduate with excellence and develop strong problem-solving and critical thinking skills for the future industry work. This design thinking project has impacted me significantly as it requires a clear problem-solving approach and solving skills. More independent study is needed for self improvement. Data engineering applications in real industries must be explored further. Overall, improvement is required to become a high potential graduate to secure future employment.

Qistina Batrisyia Binti Noor Mohd Azlan :

My dream for this course is to broaden my knowledge in data science to secure excellent industry jobs. Project timelines, team communication, soft skills and project management have been taught by this design thinking project. Coding techniques must be refined and explored to improve my potential. Lastly, ways to improve my overall practical skills to a professional standard will constantly be sought after.

6.0 CONCLUSION

Throughout this project, knowledge of design thinking phases has been acquired which helped us to clarify our vision and understanding problems to properly create an ideal solution to solve real world problems through each of its phases.

7.0 APPENDICES

Appendix A : Log Journal

Week	Description
1	The lecturer introduced the design thinking project in class. The project theme chosen was “Digital Attendance Improvement”, which focuses on how data tracking is used and students’ attendance patterns.
2	The Empathy phase was conducted using a Google Form survey and responses were collected from the students. Questions included are about problems faced and users needs.
3	The data from the Google Form were analyzed and discussed. Problems stated by respondents were difficulties scanning the attendance QR code.
4	The define phase was conducted by defining problems of the current attendance system. Every question was discussed to understand the problem faced by the users.
5	Few ideas were brainstormed to overcome attendance system problems. Each idea has been discussed to identify advantages and disadvantages and choose the best idea for our project.
6	The prototype started to be designed and created a basic flow including prototype features.
7	The testing phase is conducted using a low-fidelity prototype with users. The feedback was received and improvements needed were identified.

Appendix B : Google form survey

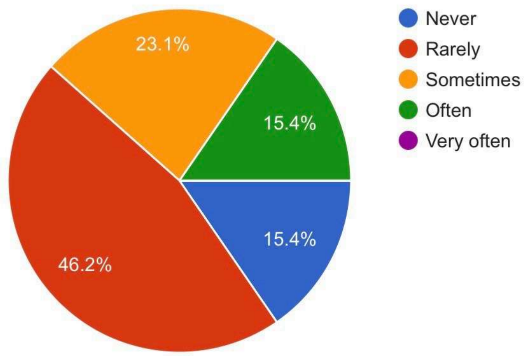


Figure 19. Frequency of Attendance Recording Problems

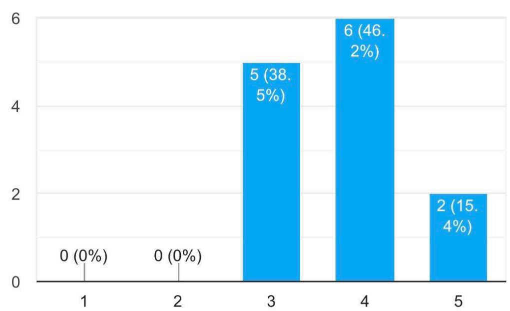


Figure 20. Rating of the Current UTM Attendance System

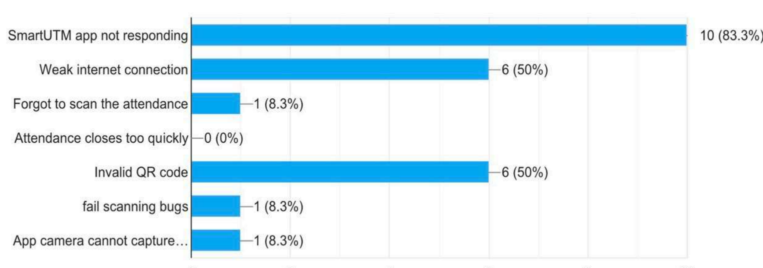
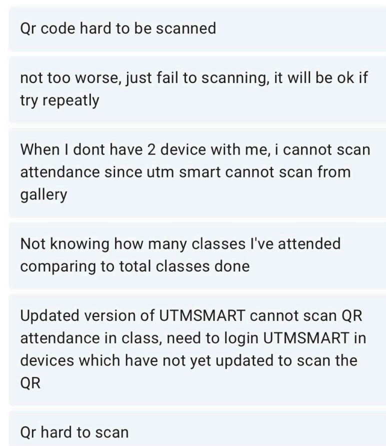


Figure 21. Common Issues in the UTM Attendance System


Qr code hard to be scanned

not too worse, just fail to scanning, it will be ok if try repeatly

When I dont have 2 device with me, i cannot scan attendance since utm smart cannot scan from gallery

Not knowing how many classes I've attended comparing to total classes done

Updated version of UTMSMART cannot scan QR attendance in class, need to login UTMSMART in devices which have not yet updated to scan the QR

Qr hard to scan

Figure 22. Worst Experiences with the Attendance System

Appendix C : List of Possible Questions and Answers

Questions	**Answers**
What issues do students face when using the QR attendance system in UTMSmart?	Students experienced issues like QR detection failure or invalid QR codes.
How does the QR system impact the efficiency of the attendance process?	Repeated errors caused more time spent.
Why does the timing of QR code displayed affect students?	QR code scanning was rushed as it was displayed only at the start of class.
How does internet connection disturb the attendance process?	Poor internet connection can fail to access the UTMSmart app.
How do students react when technical errors happen?	Signs of frustration and distress.
What happens when attendance requirements	Unable to sit for the final examination.

are not met?	

Appendix D : List of Information Gathered (Online Prototype)

No.	Aspects	Information Gathered	Reason
1.	UTM Database System	Graduate's Email Information	Graduate's Email Information will be collected to ensure attendance taken by the system is correct by comparing the present of graduate's email with the online meeting platform.
		Course Information	The student's identity will be verified under the correct assigned lecture of the course.
		Graduate's Matric ID	The graduate's matric ID is needed to prevent students from different universities entering the online meeting.
		Related Graduate's Personal Information	Related personal information such as name and age will be collected in the prototype's system.
2.	Online meeting platform	Duration In Meet	This information gathered is to notify the lecturers of students behavior in advance based on total duration entering online meetings.
		Graduate's Email Information	The correct email address is aligned with information gathered in the UTM database system.

3.	UTM Smart Apps	Duration In Meet	Lecturers will be notified about students' performance in online meetings.
		Attendant Status	The status "Successful" or "Unsuccessful" is shown to students to notify them of their attending statuses.
		Zone	The accumulated duration in meet will be calculated to showcase student's academic performance.

8.0 References

1. guide, s. (n.d.). *What Is Academic Writing? | Dos and Don'ts for Students*. Scribbr. Retrieved January 19, 2026, from https://www.scribbr.com/category/academic-writing/
2. Mortensen, D. H. (2020, July 10). *Stage 1 in the Design Thinking Process: Empathise with Your Users*. The Interaction Design Foundation. Retrieved January 19, 2026, from https://www.interaction-design.org/literature/article/stage-1-in-the-design-thinking-process-empathise-with-your-users

Link Video: https://youtu.be/VXAGALaLp0A?si=-1u7zFZQV7O42T8q